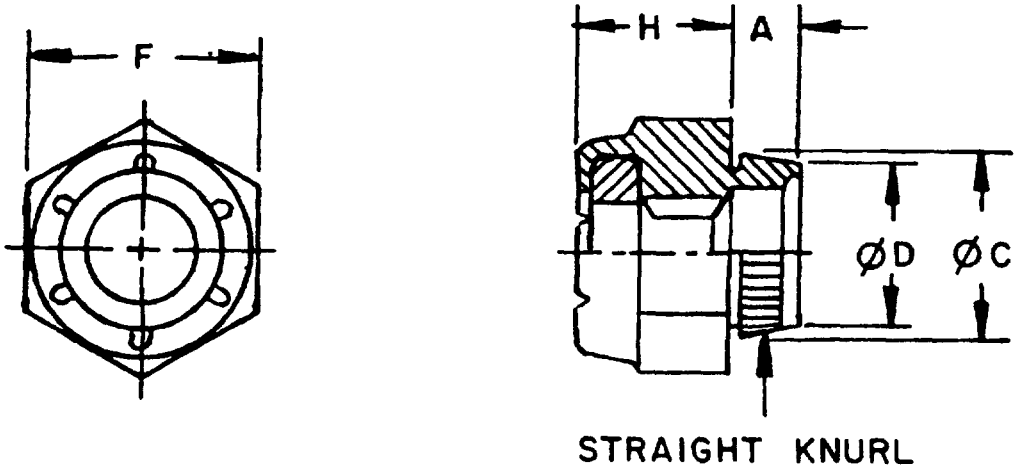
MIL-N-45938/8A
20 August 1985
SUPERSEDING
MIL-N-45938/8
30 January 1973

MILITARY SPECIFICATION SHEET

NUT, SELF-LOCKING, CLINCH (SWAGE-CLINCHING, KNURLED SHANK, 250°F, UNJC-3B)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the nuts described herein shall consist of this specification and the latest issue of MIL-N-45938.



STRAIGHT KNURL

NOTES:

1. All dimensions are in inches.

FIGURE 1. NUT, SELF-LOCKING, CLINCH (SWAGE-CLINCHING, KNURLED SHANK, 250°F, UNJC-3B).

FSC 5310

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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TABLE I. DASH NUMBERS AND DIMENSIONS.

DASH NO			THREAD SIZE	A SHANK LENGTH ±.006	SHEET THICKNESS	HOLE SIZE IN SHEET		ØC ±.002	ØD +.000 −.003	F ±.010	H ±.015
CARBON STEEL	CRES	AL ALLOY				MAX	MIN				
10	10C	10A		.063	.020 TO .031						
1	1C	1A	.112-40 UNJC-3B	.085	.032 TO .053	.186	.184	.189	.183	.250	.141
11	11C	11A		.105	.054 TO .073						
20	20C	20A		.063	.020 TO .031						
21	21C	21A		.085	.032 TO .053						
2	2C	2A	.138-32 UNJC-3B	.105	.054 TO .073	.219	.217	.222	.216	.312	.188
3	3C	3A		.135	.074 TO .103						
30	30C	30A		.165	.104 TO .133						
31	31C	31A		.195	.134 TO .163						
32	32C	32A		.085	.032 TO .053						
33	33C	33A		.105	.054 TO .073						
34	34C	34A	.164-32 UNJC-3B	.135	.074 TO .103	.271	.268	.275	.267	.375	.250
35	35C	35A		.165	.104 TO .133						
36	36C	36A		.195	.134 TO .163						
4	4C	4A		.135	.074 TO .103						
5	5C	5A	.190-24 UNJC-3B	.165	.104 TO .133	.271	.268	.275	.267	.375	.250
50	50C	50A		.195	.134 TO .163						
6	6C	6A	.250-20 UNJC-3B	.165	.104 TO .133						
7	7C	7A		.195	.134 TO .163	.355	.352	.359	.351	.438	.313

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REQUIREMENTS:

1. Material. Corrosion resistant steel, Type 303 (UNS S30300) in accordance with ASTM A581 or ASTM A582.
Aluminum alloy, Grade 2017-T4 (UNS A92017) or Grade 2014-T6 (UNS A92014) in accordance with ASTM B211.
Carbon steel, Grade 12L14 (UNS G12144) in accordance with ASTM A107, or Grade 4130 (UNS G41300) in accordance with ASTM A322, or Grade 10L18 (UNS G10180) in accordance with ASTM A510.
The locking insert shall be nylon.

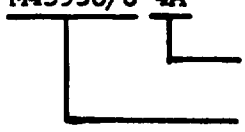
2. Protective coating and surface treatment. Aluminum alloy shall be anodized in accordance with MIL-A-8625. Carbon steel shall be cadmium plated. Corrosion resistant steel shall be cleaned, descaled, and passivated.

3. Threads. Threads shall be in accordance with MIL-S-8879 (Note : MIL-S-7742 was specified prior to MIL-N-45938/8A. Nuts with MIL-S-7742 threads are acceptable until June 1987).

4. Push-out and torque-out. Push-out and torque-out values shall be as specified in Table II when properly installed in aluminum alloy or low carbon steel sheet.

5. Part number. The part number consists of the letter "M", the basic number of this specification sheet and a dash number taken from Table I.

Example: M45938/8-4A



M45938/8-4A indicates Nut, Self-locking, Clinch (Swage-Clinching, Knurled Shank, 250°F, UNJC-3B), Aluminum alloy, .190-24 UNJC-3B, .135 Shank Length.

TABLE II. PERFORMANCE AND INSTALLATION DATA.

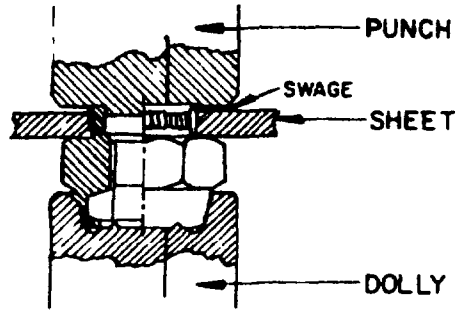
THREAD SIZE	PUSH OUT LB MIN	TORQUE-OUT IN - LB MIN	MAXIMUM INSTALLATION FORCE LB		
			AL NUT	CRES NUT	STEEL NUT
.112-40 UNJC-3B	80	10	1400	1900	1400
.138-32 UNJC-3B	120	15	1900	3000	1900
.164-32 UNJC-3B	140	20	2100	4000	2100
.190-24 UNJC-3B	200	30	2100	4000	2100
.250-20 UNJC-3B	250	50	4000	4300	4000

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RECOMMENDED INSTALLATION PROCEDURE

a. Drill or punch hole in sheet (See Table I for size).
b. Using the installation set-up shown in the sketch below, insert nut shank into hole. Select the correct Dolly from Table III below. Place the Dolly on the nut crown and, using a light hammer blow or small press, force the nut into the structure. The nut base must be fully and squarely seated against the structure.
c. Using the Dolly to support the nut head, flare the edges of the shank outward using the correct Punch from Table IV below. Do not exceed the installation forces specified in Table II.
d. Install the screw or bolt from the shank end of the nut.

INSTALLATION SET-UP



DOLLY

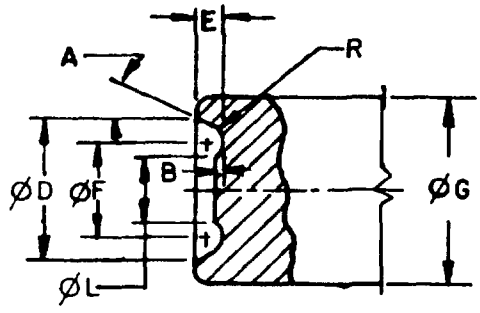


TABLE III. DOLLY-DIMENSIONS

THREAD SIZE	A ±2°	B ±.002	ØD +.002 -.000	E ±.002	ØF REF	ØG REF	ØL ±.002	R ±.001
.112-40 UNJC-3B	25°	.006	.245	.045	.173	.438	.141	.024
.138-32 UNJC-3B	25°	.010	.310	.040	.222	.438	.169	.040
.164-32 UNJC-3B	23°20'	.011	.380	.070	.247	.500	.179	.055
.190-24 UNJC-3B	15°	.008	.376	.070	.267	.500	.213	.048
.250-20 UNJC-3B	25°	.010	.451	.080	.326	.563	.273	.040

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PUNCH

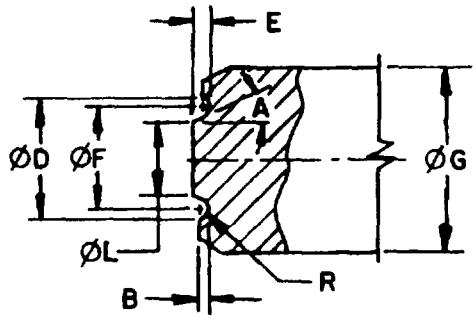


TABLE IV. PUNCH-DIMENSIONS

THREAD SIZE	A ±2°	B ±.002	ØD +.002 -.000	E ±.002	ØF REF	ØG REF	ØL ±.002	R ±.001
.112-40 UNJC-3B	25°	.008	.210	.040	.182	.438	.125	.015
.138-32 UNJC-3B	25°	.012	.258	.040	.210	.438	.136	.030
.164-32 UNJC-3B	25°	.014	.318	.038	.266	.500	.200	.023
.190-24 UNJC-3B	25°	.014	.318	.038	.266	.500	.200	.023
.250-20 UNJC-3B	30°	.012	.403	.043	.350	.563	.260	.035

Custodians:
Army - AR
Navy - OS
Air Force - 99

Review Activities:
Army - AV
DLA - IS

User Activities:
Army - AT, GL, ME, MI
Navy - AS, MC, SH, YD

Agent:
DLA - IS

Preparing Activity:
Army - AR

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